

OWEN'S PERSONALIZED

AI/ML ENGINEERING ROADMAP

9 Months · ML Engineer Track · Five Laws of Learning

Built from your Personalization Worksheet · Tailored to your goals, constraints, and strengths

Target Role	ML Engineer
Timeline	9 Months
Weekly Hours	20–40 hrs
Current Level	Basic Programming (2/5 Python)
Top Interest	NLP · Computer Vision · Recommendation Systems
Target Market	FinTech & HealthTech · Remote-First · Series A-C
Salary Target	\$120K+ / Senior ML Engineer
Law Resonated	Prediction Before Explanation

Generated June 2026 · The AI Placement Sprint · Five Laws of Learning Framework

YOUR PROFILE & PERSONALIZATION LOGIC

Student · 20-40 hrs/week	You have more time than most working professionals. Use this advantage to compress the 9-month roadmap aggressively in Phase 1.
Python 2/5 · Some NumPy/Pandas experience	You're not starting from zero, but you need to go DEEP, not just broad. Week 1 is about reading and debugging code — not writing it.
Math: Some college-level	Enough to get started. Focus on geometric intuition (3Blue1Brown) not computation. You'll pick up what you need as you apply it.
Already knows: NumPy, Pandas, Git, ML libraries	Don't restart these topics — deepen them. Move faster through Month 1 and use the saved time for extra project work.
Learning style: Break things and fix them	Every topic is structured around failure modes FIRST. Predict, break, debug, fix, post. This is your natural style — lean into it.
Biggest risk: Losing momentum at Month 3	Built-in: public checkpoints at end of Month 2 (deployed API) and Month 3 (live project). Accountability = survival.
Strengths: Communication, design, cross-domain, leadership	These are RARE in ML engineers. Your LinkedIn and GitHub will look different from everyone else. Use this from Day 1.
Target: FinTech + HealthTech, remote-first	Every project in this roadmap uses a FinTech or HealthTech dataset/use-case. Domain expertise at Month 9 = real competitive advantage.

YOUR LAW: Prediction Before Explanation — Predict first, get corrected, retain deeply. Every topic in this roadmap starts with a prediction exercise.

UNFAIR ADVANTAGES — USE THESE NOW

Advantage	How to Use It This Roadmap
Strong Communication	Post weekly LinkedIn updates. Write case studies with business impact numbers. This builds audience + proves you can explain ML to stakeholders.
Design Sense	Your GitHub READMEs will have diagrams, screenshots, and architecture visuals. Makes your work immediately stand out vs. walls of code.
Cross-Domain Translation	Reframe as your origin story: 'I approach ML like a systems thinker.' Use in cover letters, LinkedIn, and interview openers.
Leadership Skills	Position yourself as senior-track from the start. Even without experience, leaders ask the right questions — show this in interviews.

9-MONTH ROADMAP OVERVIEW

PHASE 1: FOUNDATION | Months 1–3 · 22 Topics

- Month 1: Python deep-dive + NumPy/Pandas mastery + Math (Stats, Linear Algebra, Calculus)
- Month 2: Classical ML (Linear Regression → XGBoost) + Docker + FastAPI deployment
- Month 3: Deep Learning (backprop → PyTorch → CNNs → Transformers)
- PROJECT #1: Medical image classifier deployed live on Hugging Face Spaces
- MOMENTUM CHECKPOINT: Deploy a FastAPI model by end of Month 2. Share the link publicly.

PHASE 2: COMPRESSION | Months 4–6 · 19 Topics

- Month 4: GenAI & Embeddings — Vector DBs, LLMs, RAG, Prompt Engineering, Fine-tuning
- Month 5: System Design + Advanced MLOps — Monitoring, Drift, Caching, Experiment Tracking
- Month 6: A/B Testing + Evaluation + Portfolio polish + Job search begins (Month 6)
- PROJECT #2: FinTech RAG system deployed with cost analysis (show \$15K → \$2K/month saving)
- PROJECT #3: Recommendation system with full MLOps stack (monitoring, CI/CD, drift detection)
- MOMENTUM CHECKPOINT: Start LinkedIn content series (1 post/week) by end of Month 5.

PHASE 3: OWNERSHIP | Months 7–9 · 13 Topics

- Month 7: CI/CD for ML + Security + Infrastructure as Code + Feature Engineering in Prod
- Month 8: Batch/Streaming Processing + Model Interpretability + Incident Response
- Month 9: Full job search sprint — 20+ applications/week, technical interviews, offer execution
- OFFER FRAMEWORK: Execute all 13 exercises. Target: Series A-C FinTech & HealthTech startups.
- TARGET: 2+ interviews/week by Month 9. Your 3 live projects = your portfolio. Show, don't tell.

YOUR 3 CAPSTONE PROJECTS — Built to Get Hired

These are not tutorial projects. Every project is: (1) domain-specific to FinTech or HealthTech, (2) deployed live on the internet, (3) posted publicly with metrics. Hiring managers check GitHub. These are your references.

PROJECT #1: Medical Image Classifier

Phase 1 · Month 3 · HealthTech + Computer Vision

Problem	Build a CNN-based classifier on a medical imaging dataset (chest X-ray, skin lesion, or retinal scans). Deploy as a FastAPI endpoint with a web interface.
Tech Stack	PyTorch → ResNet18 (transfer learning) → FastAPI → Docker → Hugging Face Spaces (FREE)
Metrics to Report	Test accuracy, AUC-ROC, confusion matrix. Compare transfer learning vs. scratch. Show inference time per image.
LinkedIn Post Angle	'I built a medical image classifier. Transfer learning = 92% accuracy vs 67% from scratch. Training time: 4x faster. Here's why and when to use it.'
Friction Solved	Proves Computer Vision skills + production deployment + HealthTech domain knowledge + you can ship, not just experiment

PROJECT #2: FinTech RAG Assistant

Phase 2 · Month 4 · FinTech + NLP + GenAI

Problem	Build a Retrieval-Augmented Generation (RAG) system over financial documents (SEC filings, earnings transcripts, or compliance docs). Natural language questions → grounded, cited answers.
Tech Stack	LangChain/LlamaIndex → OpenAI API (or local Ollama) → ChromaDB → FastAPI → Semantic caching layer
Metrics to Report	Retrieval recall@k, answer groundedness score, latency p50/p95, cost per query. Show cost optimization: semantic caching reduces costs by 60-80%.
LinkedIn Post Angle	'FinTech RAG at 10K queries/day: naive approach = \$15K/month. Add semantic caching + hybrid routing = \$2K/month. Here is the architecture decision.'
Friction Solved	Proves GenAI/RAG skills + NLP expertise + COST THINKING (startups care most about this) + business impact framing

PROJECT #3: ML Recommendation Engine

Phase 2 · Month 6 · Full MLOps Stack

Problem	End-to-end recommendation system (collaborative filtering + content-based hybrid) with full production stack: experiment tracking, model versioning, monitoring, A/B testing framework, and drift detection.
Tech Stack	PyTorch + scikit-learn → MLflow → FastAPI → Docker → Prometheus/Grafana → GitHub Actions CI/CD
Metrics to Report	Precision@K, Recall@K, NDCG. System metrics: p95 latency, uptime %, cost/request. Drift: detection report showing before/after data shift.
LinkedIn Post Angle	'Recommendation system: 10K requests/day, 99.2% uptime, \$0.003/request. Here is the full MLOps stack and the 3 things that broke in production.'
Friction Solved	Proves Full-stack ML Engineer + MLOps + system design thinking + operational maturity. This is the 'senior' project.

YOUR OFFER FRAMEWORK — Package Owen for Series A-C Startups

Target Market (Exercise 1)

Early-stage FinTech and HealthTech startups, Series A-C, 50-300 employees, US-funded, remote-first. Why demand is strong: (1) FinTech needs ML urgently for fraud detection, credit scoring, and compliance automation. (2) HealthTech needs explainable AI for FDA compliance and patient risk — underserved by generic ML engineers. (3) Neither can afford a \$200K senior engineer but both desperately need ML capability to compete. How to find them: LinkedIn advanced search (company size 50-300, stage Series A-C), AngelList, Y Combinator batch lists, TechCrunch funding announcements.

Dream Outcome (Exercise 2) — What They Actually Want

GET: Production-ready ML features generating measurable business impact (fraud reduction %, diagnostic accuracy %, recommendation CTR improvement) IN: 60-90 days (before the next board meeting or funding milestone) WITHOUT: Hiring a \$200K senior engineer OR spending 3 months onboarding someone who cannot ship YOUR PROMISE: 'Get a deployed, monitored ML system generating measurable results in 60-90 days, without the cost or risk of a senior hire.' YOUR COMMUNICATION EDGE: You can explain ML value to non-technical founders. Most ML engineers cannot do this. This is your #1 differentiator beyond code.

Hiring Frictions and Solutions (Exercises 4 & 5)

FRICION 1: No prior ML work experience SOLUTION: 3 deployed projects with real metrics. 'I built X, it handles Y requests/day, it reduced cost by Z%.' Work > credentials. FRICION 2: No portfolio of real projects SOLUTION: Projects #1, #2, #3 — all live on the internet, all with metrics, all in FinTech or HealthTech domains. FRICION 3: Do not understand business context of ML SOLUTION: Cost case study blog post: 'Reduced LLM inference cost from \$15K to \$2K/month with semantic caching.' Shows ROI thinking. FRICION 4: Career gap / transitioning fields SOLUTION: Reframe as advantage. 'I approach ML problems with cross-domain systems thinking — rare in pure ML engineers.' FRICION 5: No professional software development experience SOLUTION: CI/CD pipeline, 99%+ uptime logs, incident runbooks. Show operational maturity, not just notebooks.

Your Offer & Delivery Model (Exercises 6-13)

AVAILABILITY: Full-time immediately. Student — can start any date. TRIAL OFFER: 'Assess my work on a 30-day paid trial. If I do not ship something production-ready by Day 30, you owe me nothing.' This removes their biggest risk: a wrong hire. WEEK 1 CONTRIBUTION: Audit current ML stack, identify 3 quick cost or quality wins, deliver improvement proposal. YOUR OFFER NAME OPTIONS: 1. 'The ML-to-Production Engineer who speaks business and code' 2. 'Full-stack ML Engineer: from model to monitoring in 60 days' 3. 'The FinTech/HealthTech ML Engineer who ships, not just experiments' LINKEDIN ABOUT SECTION (use this now): 'ML Engineer building production AI systems for FinTech and HealthTech startups. I design, deploy, and monitor end-to-end ML pipelines — from RAG systems to computer vision classifiers. ML value = shipped model + business metric improvement. Currently building in public. Seeking remote ML Engineer roles at mission-driven Series A-C companies.'

MOMENTUM PROTECTION SYSTEM

YOUR BIGGEST RISK: 'I will push 3 months in and lose the momentum along the way.' This roadmap is built specifically to prevent that. Here is your system.

END OF WEEK 4

Post your first LinkedIn analysis — a data cleaning project or stats notebook. Public commitment is the most powerful accountability mechanism. Once you post, your network knows. That creates positive pressure to keep going.

END OF MONTH 2

CHECKPOINT: Deploy something. Even a simple FastAPI model on your laptop via ngrok counts. The act of shipping — even small — rewires your brain. You become 'someone who ships' instead of 'someone who is learning.' This identity shift is crucial for Month 3.

END OF MONTH 3 (HIGH RISK ZONE)

CHECKPOINT: Project #1 live on the internet with a URL. Post it on LinkedIn. This is the hardest transition — from foundation to application. Celebrate the milestone. Write a post about what you learned in 3 months. Publicly reflecting = momentum fuel.

MONTH 3 TO MONTH 4

The danger zone. Foundation is done; GenAI feels overwhelming. STRATEGY: On Day 1 of Month 4, re-read your Month 3 LinkedIn posts. Look at how far you have come. Then apply Prediction Before Explanation to your first GenAI topic — predict how embeddings work before reading a single article. Curiosity is momentum.

END OF MONTH 5

CHECKPOINT: Start LinkedIn content series — 1 post per week minimum from here on. Content strategy: 'What I learned about [topic] this week + the failure mode I hit + how I fixed it.' This builds your audience WHILE you learn, so by Month 9 you have warm leads, not cold applications.

MONTH 6 ONWARD

Apply to 3-5 companies per week while learning. Treat every rejection as data — update your pitch, not your confidence. Continue shipping projects. Continue posting. The compound effect of 3 months of public building + technical skills is what makes you the obvious choice.

FINAL COMMITMENT: Q36 — 'The only thing that could make me quit is laziness — otherwise I am locked in.' This roadmap assumes you meant it. Every checkpoint above is a mechanism to ensure laziness never wins.